Problem Set 3

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PUBLISHED
November 2, 2026

GitHub repo: https://github.com/Noskcaj8/ec422_ps3_Dixon

```
library(tidyverse)
library(fpp3)

retail_raw <- read_csv(
  "data/retail_sales_raw.csv",
  na = ".",
  col_types = cols(
    observation_date = col_date(format = "%Y-%m-%d"),
    RSAFS            = col_double(),
    RSAFSNA          = col_double()
  )
)

retail_raw_long <- retail_raw |>
  select(observation_date, RSAFS, RSAFSNA) |>
  pivot_longer(cols = c(RSAFS, RSAFSNA),
               names_to = "series_id", values_to = "sales")

retail_ts <- retail_raw_long |>
  mutate(Month = yearmonth(observation_date)) |>
  as_tsibble(index = Month, key = series_id)

can_gas <- canadian_gas |> select(Month, Volume)
```

Time and Tsibbles

Datasets. `retail_ts` holds two monthly U.S. retail and food services sales series from the Census Bureau, downloaded from FRED: RSAFS (seasonally adjusted) and RSAFSNA (not seasonally adjusted), in millions of dollars. Its index is `Month` (a `yearmonth`) and its key is `series_id`. `can_gas` is monthly Canadian natural gas production in billions of cubic meters, January 1960 to February 2005, from the `fpp3` package. It has a single series, so it has no key. Its index is `Month`, and the variable is `Volume`.

Reading the data (2.2-2.4)

```
head(retail_raw)
```

```
# A tibble: 6 × 3
  observation_date  RSAFS RSAFSNA
  <date>            <dbl>   <dbl>
1 1992-01-01       159177  142051
2 1992-02-01       159189  142498
3 1992-03-01       158647  154439
4 1992-04-01       159921  158430
5 1992-05-01       160471  164788
6 1992-06-01       161205  163417
```

The `observation_date` column is read as `<date>`. (Default `read_csv()` guessed correctly here, but I specify column types explicitly anyway.)

```
head(retail_raw_long)
```

```
# A tibble: 6 × 3
  observation_date series_id  sales
  <date>           <chr>      <dbl>
1 1992-01-01       RSAFS     159177
2 1992-01-01       RSAFSNA   142051
3 1992-02-01       RSAFS     159189
4 1992-02-01       RSAFSNA   142498
5 1992-03-01       RSAFS     158647
6 1992-03-01       RSAFSNA   154439
```

```
head(retail_ts)
```

```
# A tsibble: 6 x 4 [1M]
# Key:       series_id [1]
  observation_date series_id  sales    Month
  <date>           <chr>      <dbl>    <mth>
1 1992-01-01       RSAFS     159177 1992 Jan
2 1992-02-01       RSAFS     159189 1992 Feb
3 1992-03-01       RSAFS     158647 1992 Mar
4 1992-04-01       RSAFS     159921 1992 Apr
5 1992-05-01       RSAFS     160471 1992 May
6 1992-06-01       RSAFS     161205 1992 Jun
```

```
has_gaps(retail_ts)
```

```
# A tibble: 2 × 2
  series_id .gaps
  <chr>     <lgl>
1 RSAFS     FALSE
```

```
2 RSAFSNA    FALSE
```

The tsibble header shows `[1M]`, meaning monthly frequency was detected, and `has_gaps()` is `FALSE` for both series.

Canadian gas (2.5)

```
key_vars(canadian_gas)
```

```
character(0)
```

```
index_var(canadian_gas)
```

```
[1] "Month"
```

```
has_gaps(can_gas)
```

```
# A tibble: 1 × 1
  .gaps
  <lgl>
1 FALSE
```

```
head(can_gas)
```

```
# A tsibble: 6 x 2 [1M]
     Month Volume
     <mth>  <dbl>
1 1960 Jan   1.43
2 1960 Feb   1.31
3 1960 Mar   1.40
4 1960 Apr   1.17
5 1960 May   1.12
6 1960 Jun   1.01
```

Working with dates (2.6)

```
can_gas_recent <- can_gas |>
  filter(Month >= yearmonth("1990 Jan")) |>
  mutate(year = year(Month),
         month = month(Month, label = TRUE),
         quarter = quarter(Month))

can_gas_recent |>
  as_tibble() |>
  group_by(month) |>
```

```
  summarize(avg_volume = mean(Volume)) |>
  knitr::kable(digits = 1)
```

month	avg_volume
Jan	17.0
Feb	15.3
Mar	16.3
Apr	15.7
May	15.7
Jun	14.9
Jul	15.6
Aug	15.6
Sep	15.1
Oct	16.2
Nov	16.2
Dec	17.1

Since 1990, December has the highest average production (17.1, with January close at 17.0) and June has the lowest (about 14.9). Production is higher in winter and lower in summer.

```
can_gas_recent |>
  mutate(next_month = Month + 1,
         same_month_last_year = Month - 12) |>
  head()
```

```
# A tsibble: 6 x 7 [1M]
     Month Volume  year month quarter next_month
same_month_last_year
     <mth>  <dbl> <dbl> <ord>   <int>      <mth>
<mth>
1 1990 Jan   12.5  1990 Jan         1   1990 Feb
1989 Jan
2 1990 Feb   11.2  1990 Feb         1   1990 Mar
1989 Feb
3 1990 Mar   11.6  1990 Mar         1   1990 Apr
1989 Mar
4 1990 Apr   11.1  1990 Apr         2   1990 May
1989 Apr
5 1990 May   11.0  1990 May         2   1990 Jun
1989 May
6 1990 Jun   10.1  1990 Jun         2   1990 Jul
```

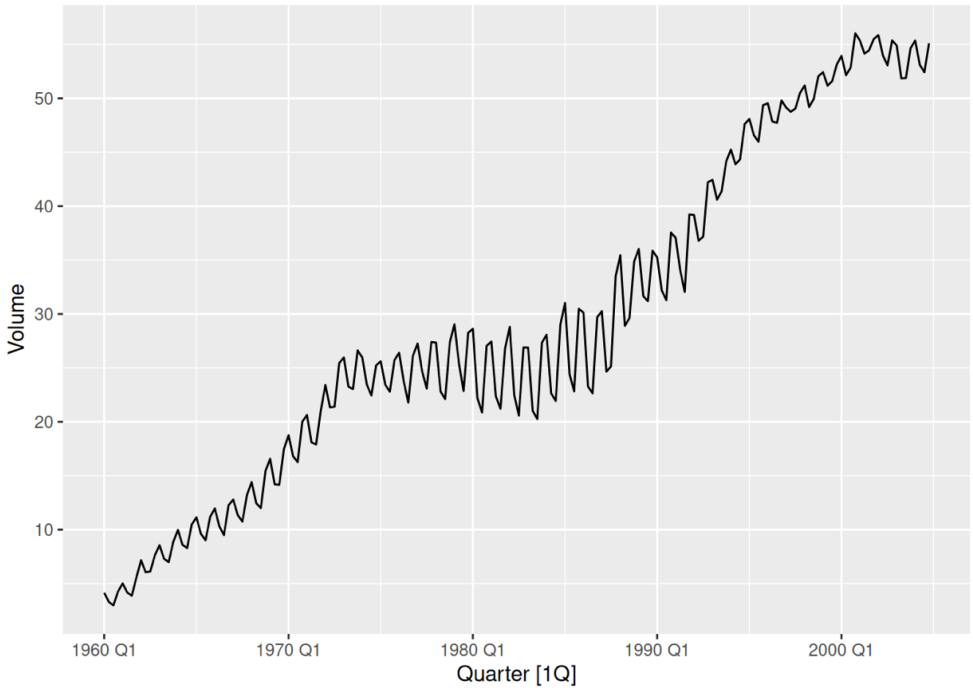
```
1989 Jun
```

```
can_gas_q <- can_gas |>
  filter(Month <= yearmonth("2004 Dec")) |>
  index_by(Quarter = yearquarter(Month)) |>
  summarize(Volume = sum(Volume))

head(can_gas_q)
```

```
# A tsibble: 6 x 2 [1Q]
  Quarter Volume
    <qtr>  <dbl>
1 1960 Q1   4.14
2 1960 Q2   3.30
3 1960 Q3   2.97
4 1960 Q4   4.26
5 1961 Q1   5.01
6 1961 Q2   4.17
```

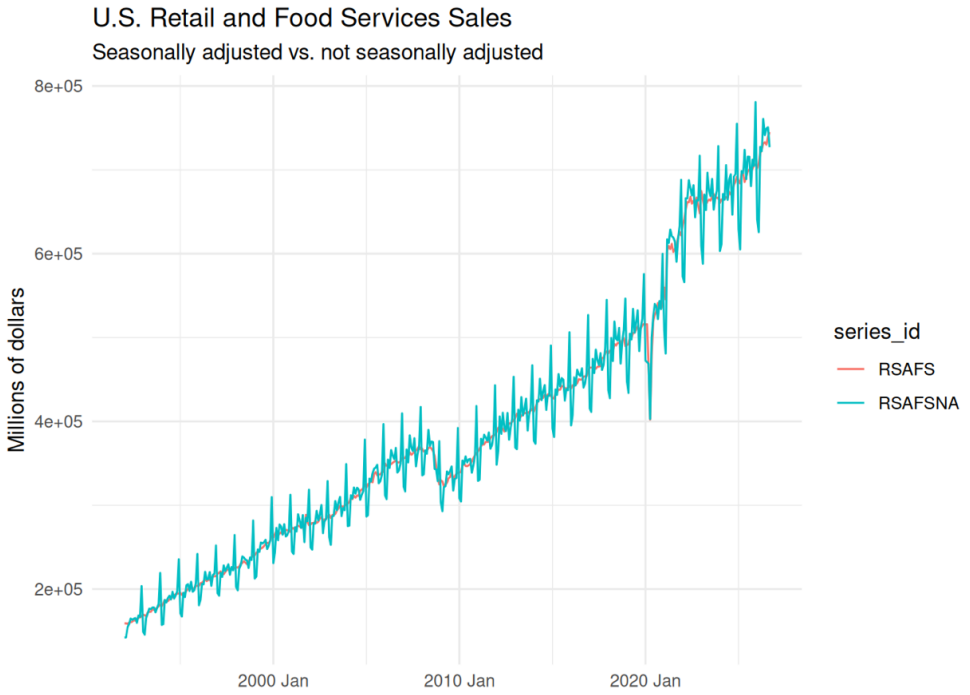
```
can_gas_q |> autoplot(Volume)
```



I used `sum()` because gas production is a flow: quarterly production should be the total produced over the three months, not the average monthly rate. The quarterly series is still clearly seasonal, with a repeating yearly pattern of winter highs and summer lows.

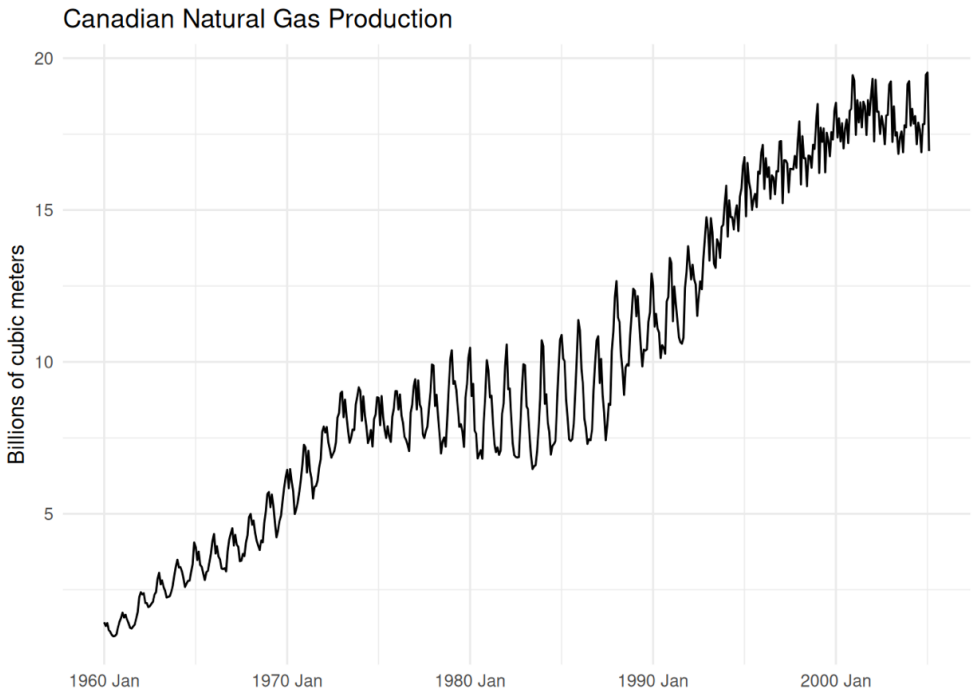
Plotting (2.7)

```
retail_ts |>
  autoplot(sales) +
  labs(title = "U.S. Retail and Food Services Sales",
       subtitle = "Seasonally adjusted vs. not seasonally adjus
       x = NULL, y = "Millions of dollars", color = NULL) +
  theme_minimal()
```

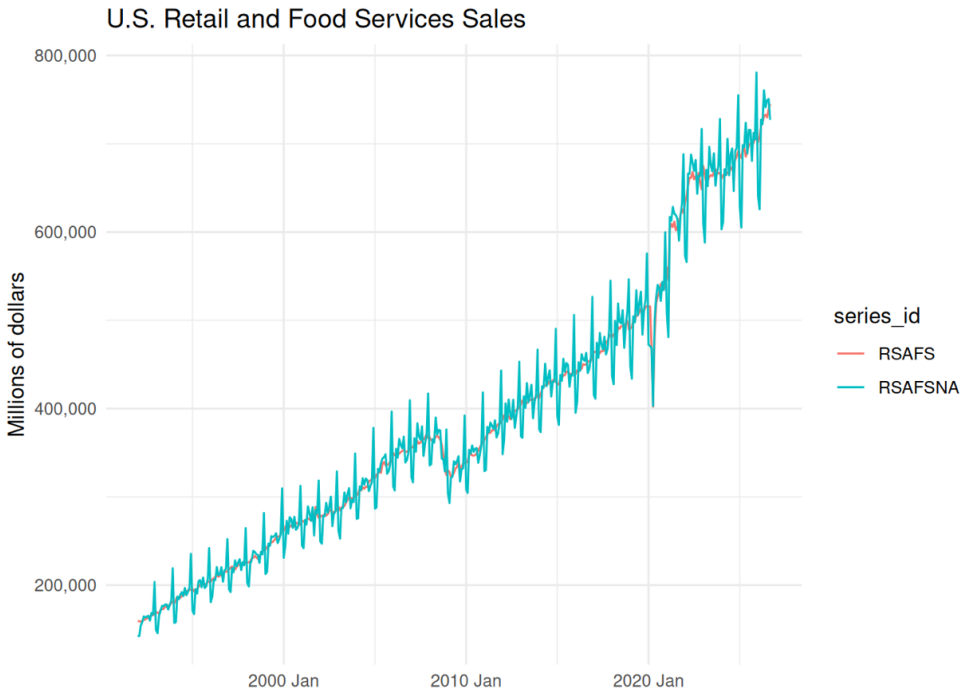


The NSA series has a sharp spike and dip that repeat every year. Probably due to the holiday shopping surge near the end of the year, followed by a drop afterword. The SA series doesn't show this, much smoother because the seasonal adjustment got rid of that repeating in year pattern. In the long run both series follow the same path. Both rise steadily over time, same drop in 2020 (covid recession). Seasonal adjustment removes the predictable in year fluctuation, but leaves the trend and the major shocks intact.

```
can_gas |>
  autoplot(Volume) +
  labs(title = "Canadian Natural Gas Production",
       x = NULL, y = "Billions of cubic meters") +
  theme_minimal()
```




```
retail_ts |>
  autoplot(sales) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(title = "U.S. Retail and Food Services Sales",
       x = NULL, y = "Millions of dollars", color = NULL) +
  theme_minimal()
```

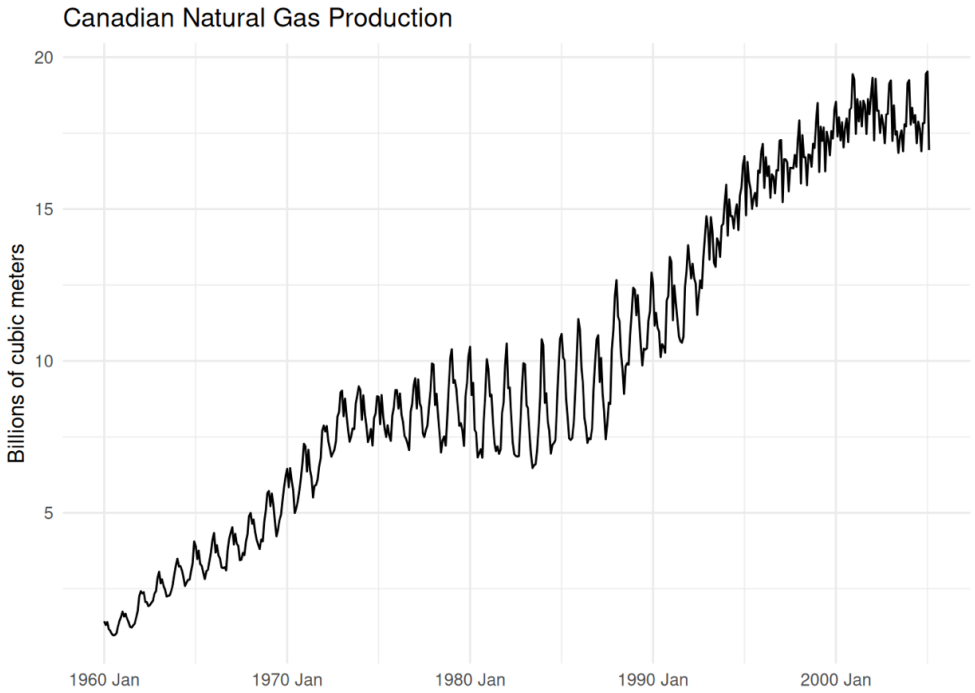


Transformations

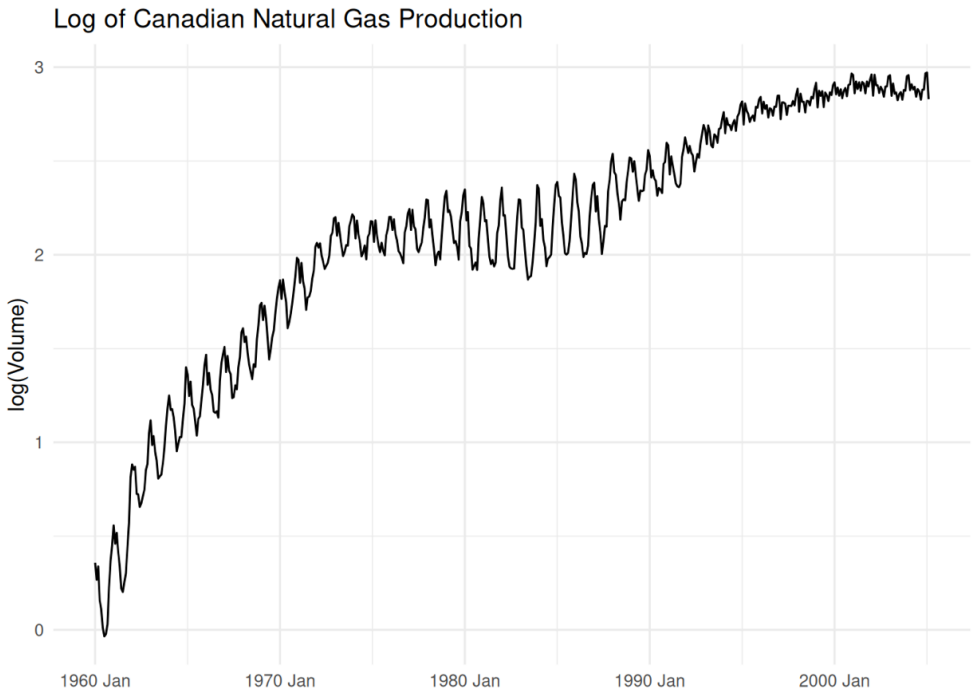
Logs (3.1)

```
can_gas <- can_gas |>
  mutate(log_volume = log(Volume))

can_gas |>
  autoplot(Volume) +
  labs(title = "Canadian Natural Gas Production",
       x = NULL, y = "Billions of cubic meters") +
  theme_minimal()
```




```
can_gas |>
  autoplot(log_volume) +
  labs(title = "Log of Canadian Natural Gas Production",
       x = NULL, y = "log(Volume)") +
  theme_minimal()
```



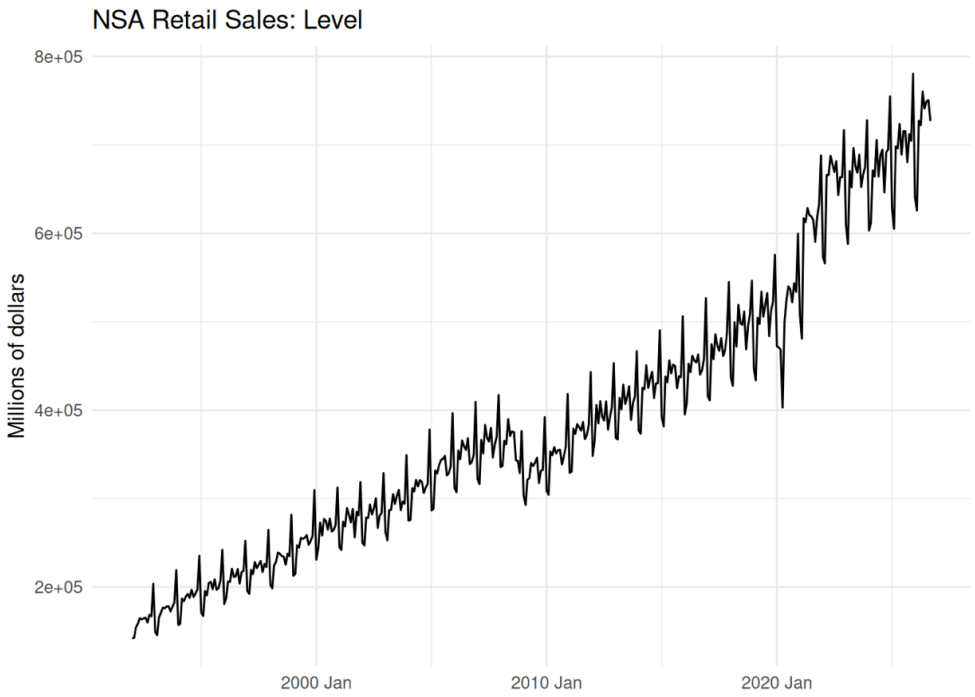
3.1a. For the first part of the sample, the seasonal swings do seem to grow with the level of the series. In the 60s when production is low, the yearly swings are small. By the 70s and 80s, when the level has risen to 8-10, the swings are much bigger. This doesn't hold over the entire sample though. After 1990, the series keeps climbing, but the swings don't grow proportionally. They look similar in size if not smaller, and the seasonal pattern becomes less regular. The swings grow with the series early on, then the relationship breaks down. Swings are largest in the 1980s, then later swings get smaller.

3.1b. After taking logs, the trend no longer looks like steady growth. It rises steeply through the 60s and early 70s, then flattens out through the 80s, and rises again in the 90s, then tapers off. The seasonal swings are more even through the first half of the sample than in the raw series, but they now obviously shrink in the 90s and 2000s. I think logs are only a partial help for this because they tame the pattern where swings grew with the level, but the trend still isn't a clean straight line, and now the later swings look too small. So I'd use logs, but I don't expect them to fix everything.

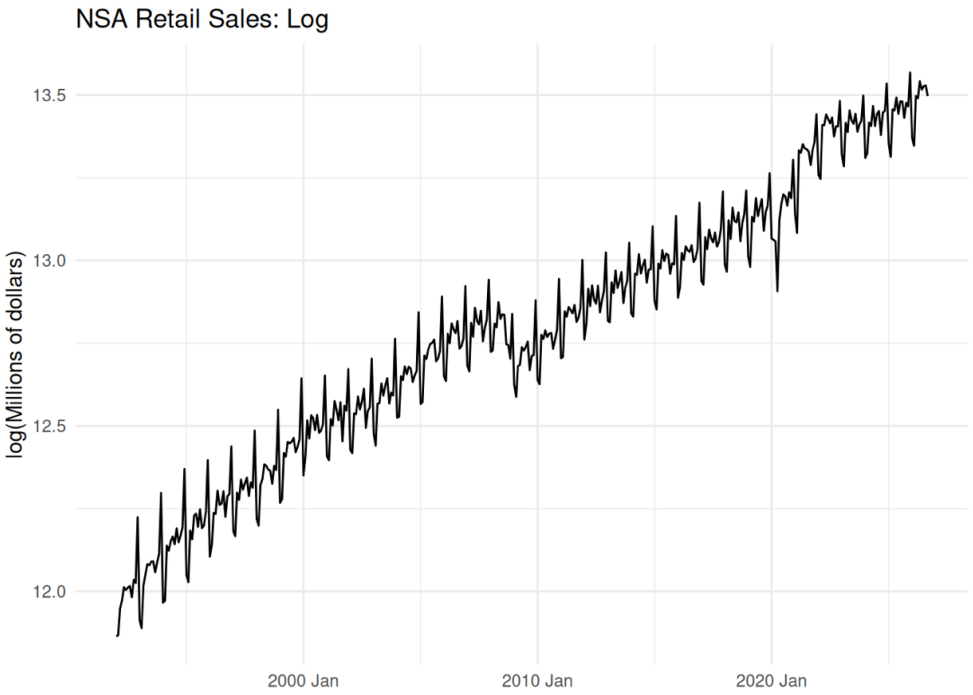
```
retail_nsa <- retail_ts |>
  filter(series_id == "RSAFSNA") |>
  mutate(log_sales = log(sales))

retail_nsa |>
  autoplot(sales) +
```

```
  labs(title = "NSA Retail Sales: Level",
       x = NULL, y = "Millions of dollars") +
  theme_minimal()
```




```
retail_nsa |>
  autoplot(log_sales) +
  labs(title = "NSA Retail Sales: Log",
       x = NULL, y = "log(Millions of dollars)") +
  theme_minimal()
```

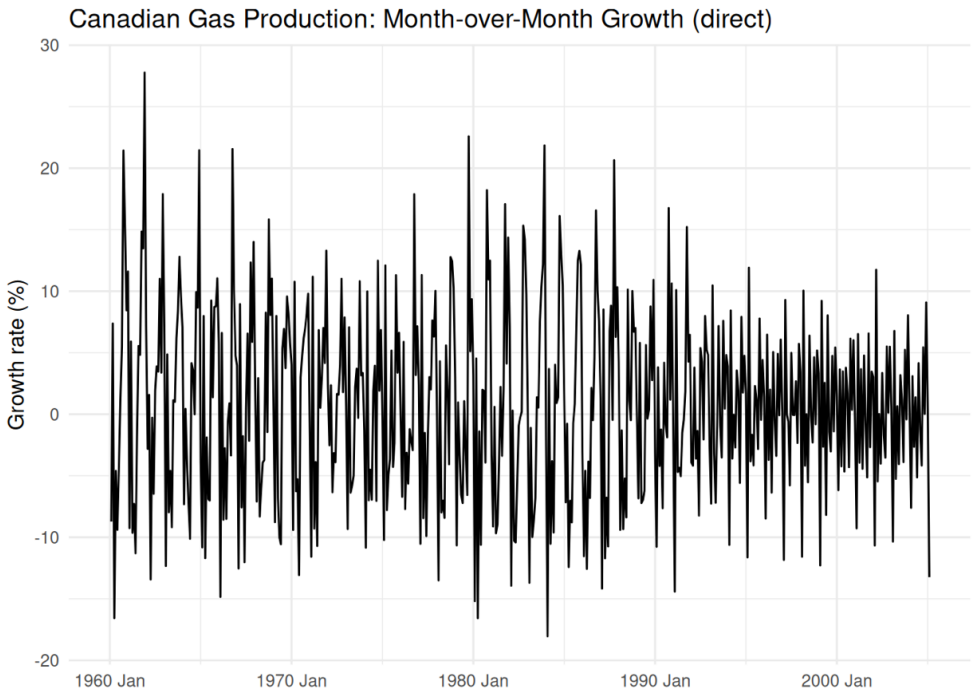


3.1c. The case for logging is stronger for retails than for gas. In levels, the seasonal spikes get bigger as sales grow. In logs, the trend is close to a straight line and the seasonal swings are fairly constant in size across the entire sample. Retail sales grow with at a pretty steady percentage rate, which is what logs turn into a straight line.

Growth rates (3.2)

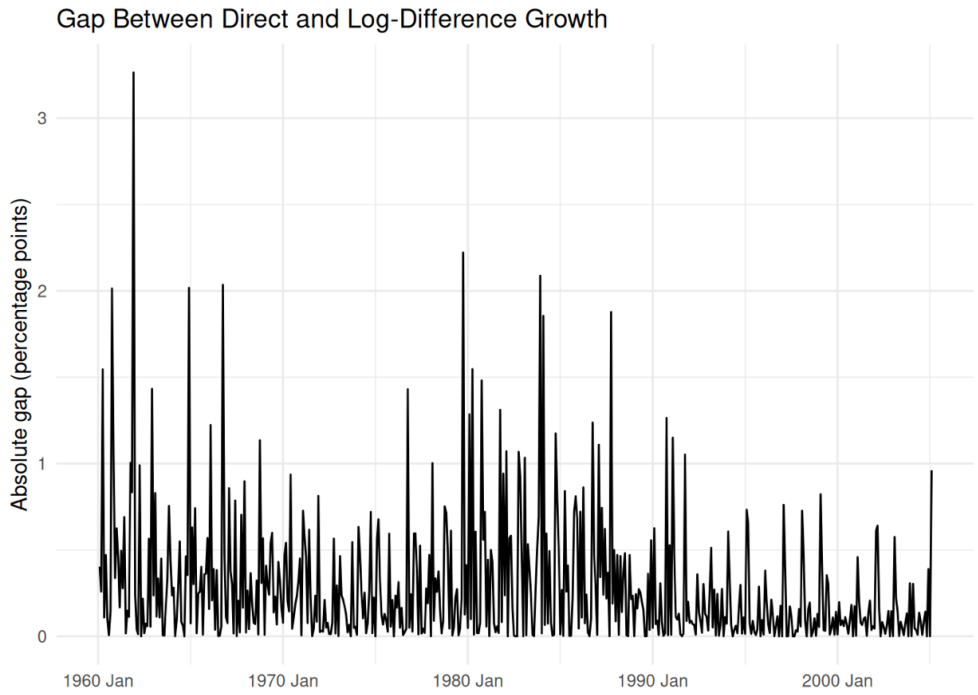
```
can_gas <- can_gas |>
  mutate(
    growth_direct  = 100 * difference(Volume) / lag(Volume),
    growth_logdiff = 100 * difference(log_volume)
  )

can_gas |>
  autoplot(growth_direct) +
  labs(title = "Canadian Gas Production: Month-over-Month Growt
       x = NULL, y = "Growth rate (%)") +
  theme_minimal()
```




```
can_gas <- can_gas |>
  mutate(growth_gap = abs(growth_direct - growth_logdiff))

can_gas |>
  autoplot(growth_gap) +
  labs(title = "Gap Between Direct and Log-Difference Growth",
       x = NULL, y = "Absolute gap (percentage points)") +
  theme_minimal()
```

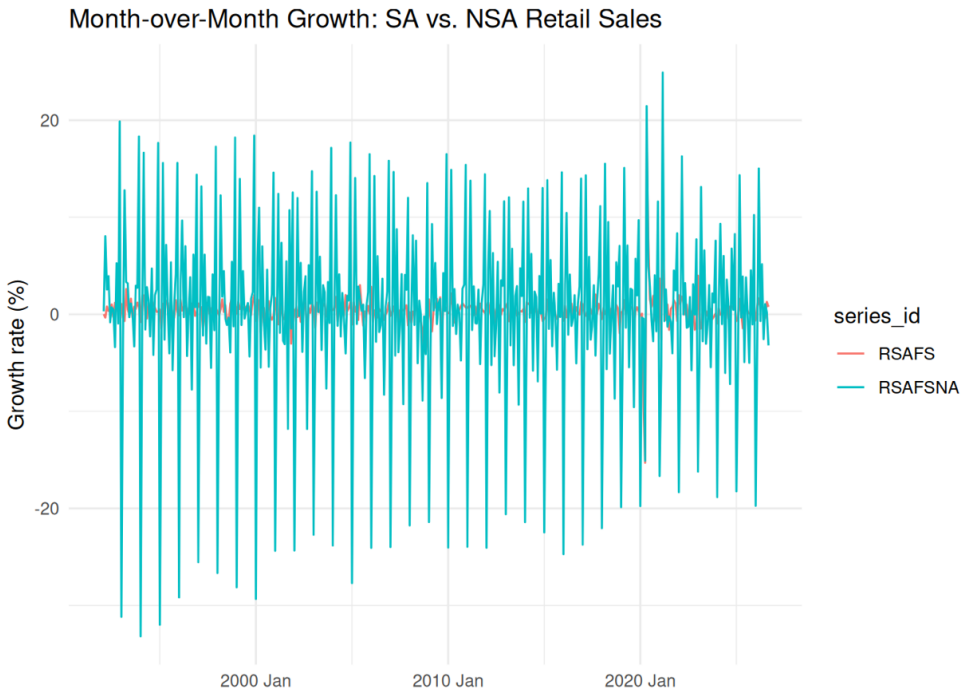


3.2a. The log difference approximation is very good for most of the sample. The gap is usually well under 1 percentage point. It's largest in the 60s-80s (peaks at 2-3 points), when monthly growth rates were large, and shrinks in the later years when growth rates are smaller. That's the usual pattern, since the approximation gets worse as growth rates get bigger.

SA vs. NSA growth rates (3.3)

```
retail_ts <- retail_ts |>
  group_by_key() |>
  mutate(growth_mom = 100 * difference(log(sales))) |>
  ungroup()

retail_ts |>
  autoplot(growth_mom) +
  labs(title = "Month-over-Month Growth: SA vs. NSA Retail Sale
       x = NULL, y = "Growth rate (%)", color = NULL) +
  theme_minimal()
```

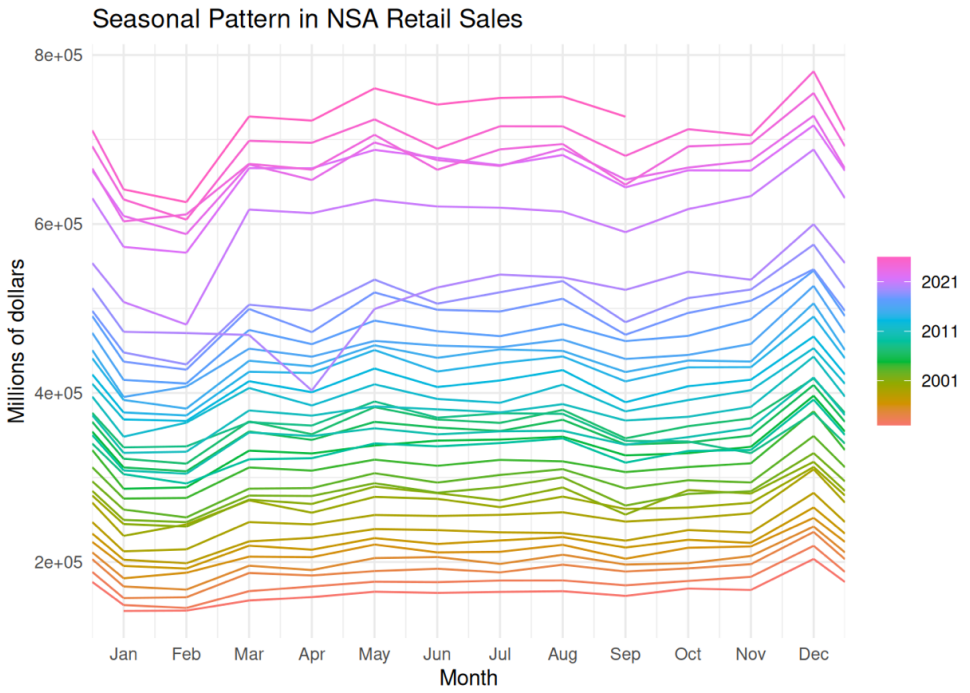


3.3. The NSA series is much noisier. It's month over month growth swings between roughly -20% and 20% while the SA series stays close to zero with very small movements. The reason being that the NSA series still contains the repeating seasonal pattern so its month-to-month changes are dominated by the seasonal movement. Seasonal adjustment removes that pattern so the SA growth rates only reflect the underlying changes in the economy.

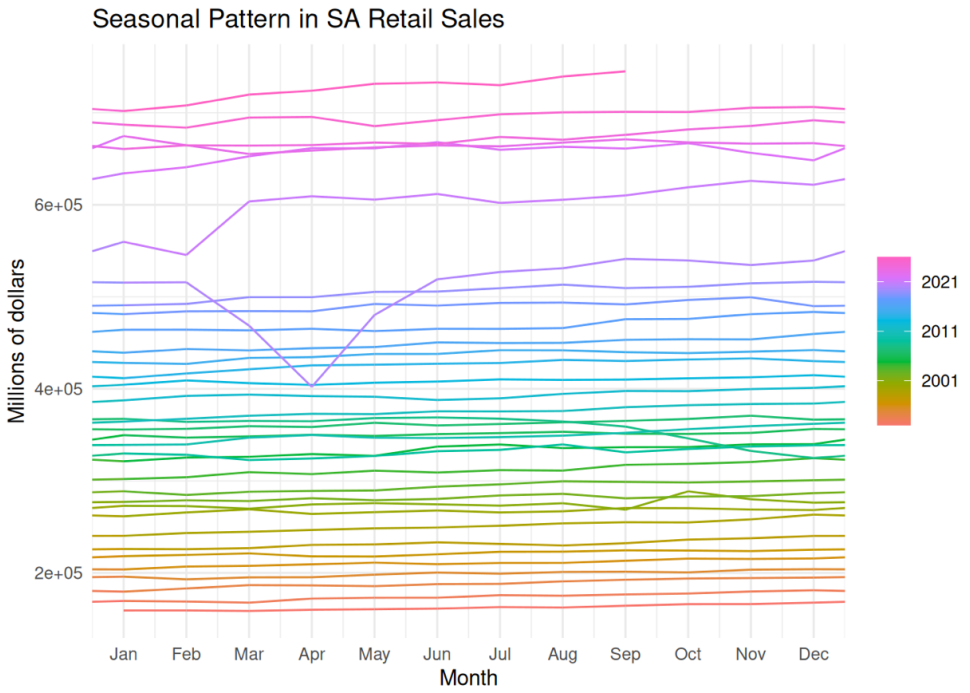
Seasonality

Season plots (4.1)

```
retail_ts |>
  filter(series_id == "RSAFSNA") |>
  gg_season(sales) +
  labs(title = "Seasonal Pattern in NSA Retail Sales",
       y = "Millions of dollars") +
  theme_minimal()
```

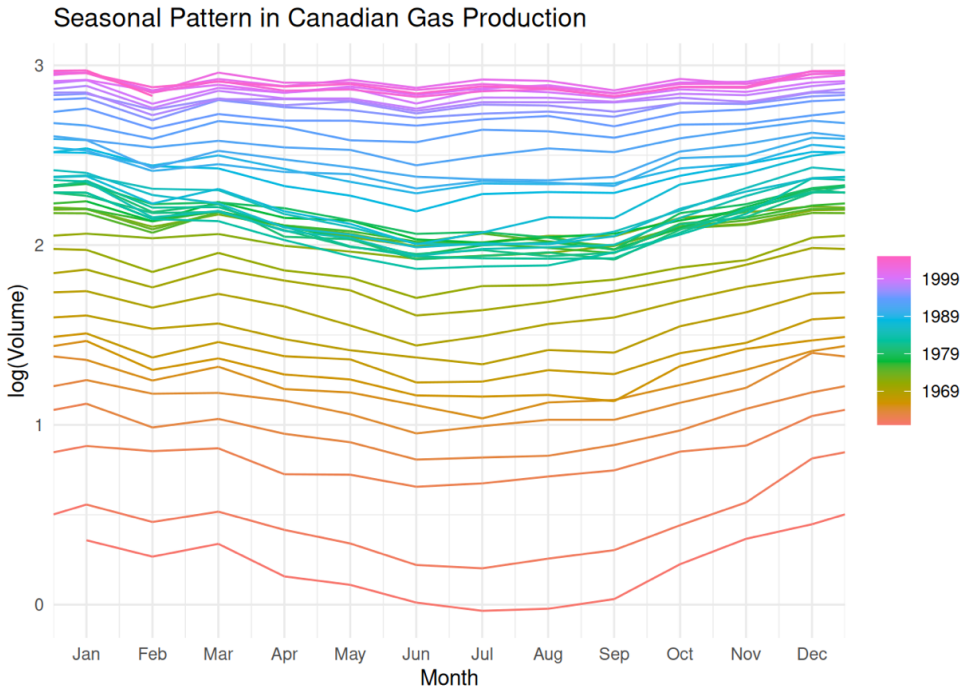



```
retail_ts |>
  filter(series_id == "RSAFS") |>
  gg_season(sales) +
  labs(title = "Seasonal Pattern in SA Retail Sales",
       y = "Millions of dollars") +
  theme_minimal()
```



4.1b. In the NSA plot, the yearly lines are almost identical and trace the same shape. A dip in February, a jump in March, a flat stretch through summer, a dip in September, and a climb to the highest point in December. The pattern repeats year after year with only the level shifting upward. In the SA plot, that shared shape is gone and the lines are nearly flat across the months. The one exception being Covid. This shows seasonal adjustment in action, estimating the repeating in the NSA plot then removing it.

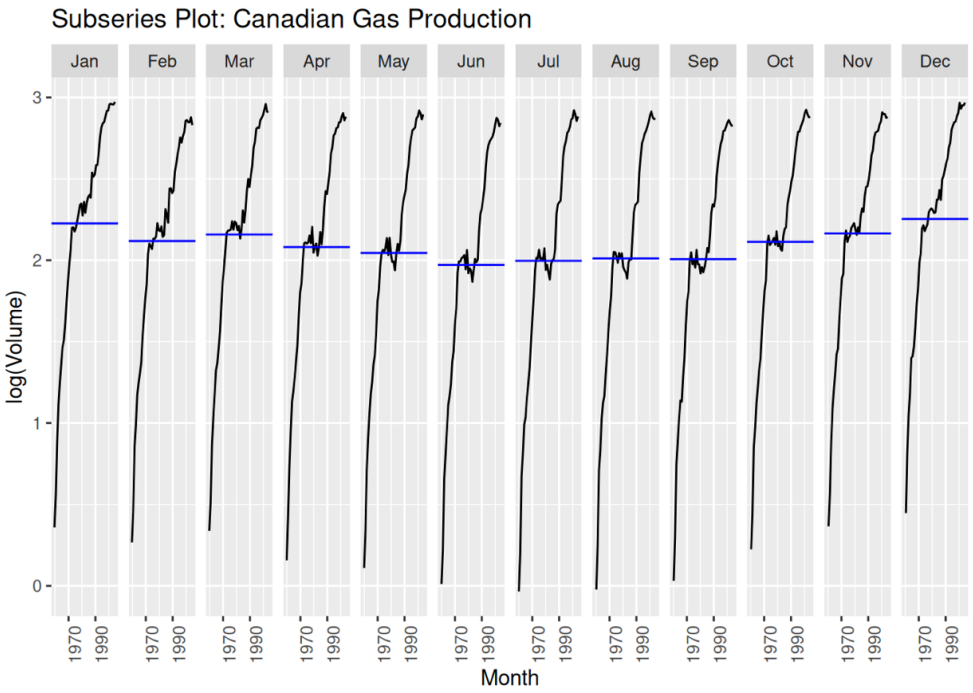
```
can_gas |>
  gg_season(log_volume) +
  labs(title = "Seasonal Pattern in Canadian Gas Production",
       y = "log(Volume)") +
  theme_minimal()
```



4.1c. Production is highest in the winter months, and lowest in summer. So the lines form a shallow U-shape. This makes sense because natural gas is used heavily for heating so demand rises in cold months.

Subseries plot (4.2)

```
can_gas |>
  gg_subseries(log_volume) +
  labs(title = "Subseries Plot: Canadian Gas Production",
       y = "log(Volume)")
```

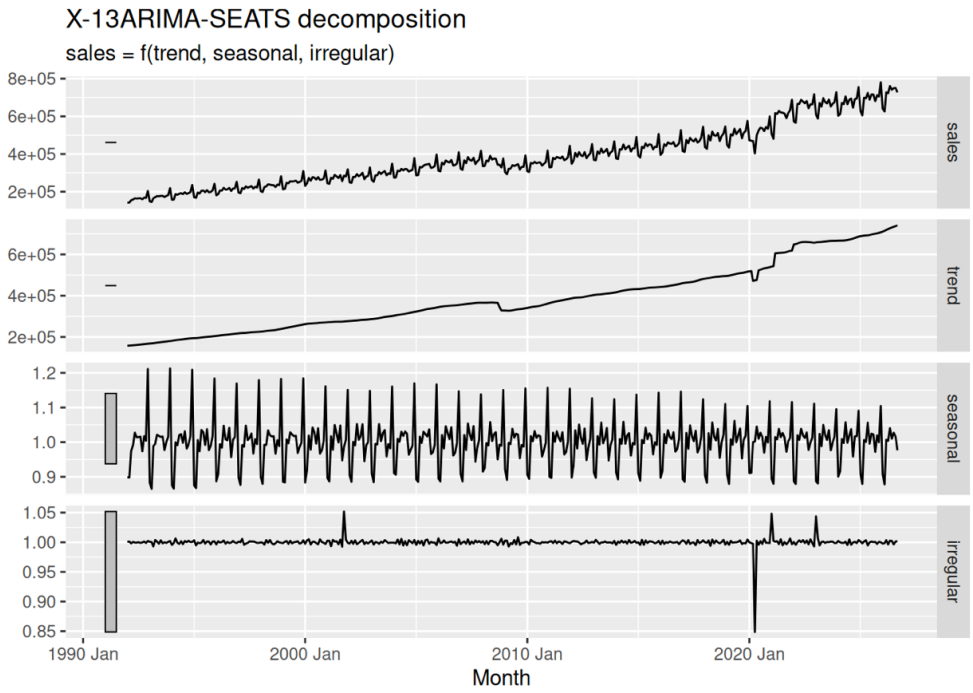


The blue mean lines are highest in December and January and lowest in summer months confirming the season plot. Every month's panel shows the same strong upward trend over time, so the seasonal pattern is stable as the level rises.

X-13ARIMA-SEATS (4.3)

```
retail_nsa_x13 <- retail_ts |>
  filter(series_id == "RSAFSNA") |>
  model(x13 = X_13ARIMA_SEATS(sales)) |>
  components()

retail_nsa_x13 |> autoplot()
```



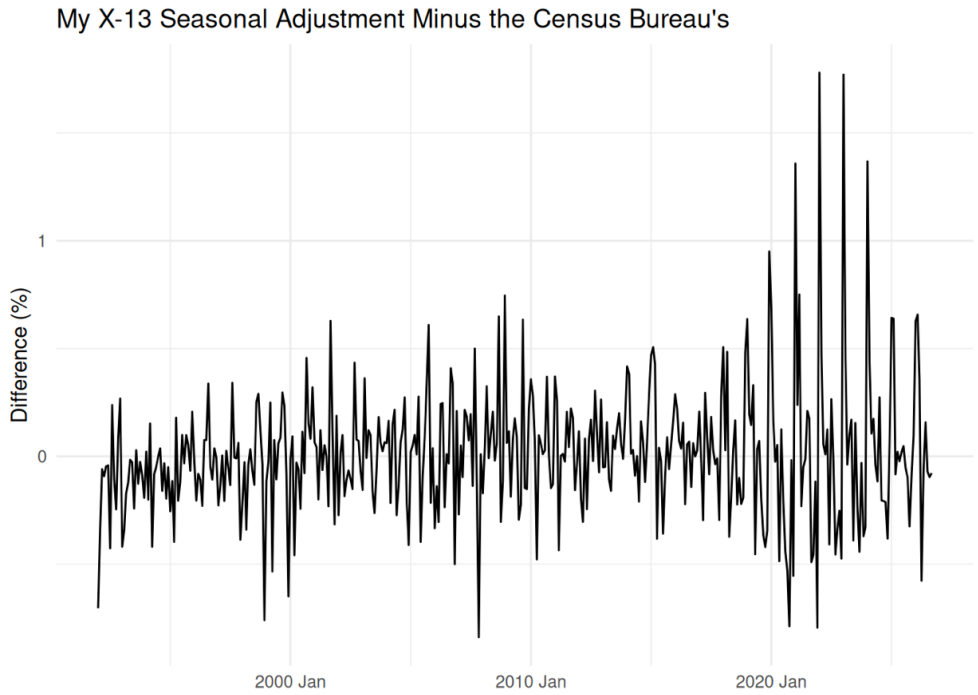

```
retail_nsa_x13 |>
  as_tibble() |>
  mutate(mon = month(Month, label = TRUE)) |>
  group_by(mon) |>
  summarize(avg_seasonal_factor = mean(seasonal)) |>
  knitr::kable(digits = 3)
```

mon	avg_seasonal_factor
Jan	0.899
Feb	0.891
Mar	1.009
Apr	0.992
May	1.043
Jun	1.013
Jul	1.013
Aug	1.030
Sep	0.964
Oct	0.994
Nov	1.004
Dec	1.148

4.3a. The seasonal factor is highest in December, at about 1.15. Sales in December are typically about 15% above the trend. The lowest months are January and February (about 0.9 and 0.89, so roughly 10% below trend). The seasonal panel looks fairly stable, with no obvious growth in the swings. This fits a multiplicative decomposition, where the seasonal effect is a constant percentage of the trend rather than a constant of dollar amount.

```
sa_compare <- retail_nsa_x13 |>
  as_tibble() |>
  select(Month, mine = season_adjust) |>
  left_join(
    retail_ts |>
      filter(series_id == "RSAFS") |>
      as_tibble() |>
      transmute(Month, census = sales),
    by = "Month"
  ) |>
  mutate(pct_diff = 100 * (log(mine) - log(census)))

sa_compare |>
  ggplot(aes(x = Month, y = pct_diff)) +
  geom_line() +
  labs(title = "My X-13 Seasonal Adjustment Minus the Census Bu
       x = NULL, y = "Difference (%)") +
  theme_minimal()
```



4.3b. My X-13 adjustment is very close to the official one. The difference is centered on zero and mostly stays within about 1 percentage point. The differences are largest in the 2020s but that might be due to the Census Bureau using outlier treatments for Covid while I used the defaults.

Linear trend + seasonal dummies (4.4)

```
can_gas_fit <- can_gas |>
  model(TSLM(log_volume ~ trend() + season()))

can_gas_fit |>
  tidy() |>
  select(term, estimate, std.error, statistic, p.value) |>
  knitr::kable(digits = 3)
```

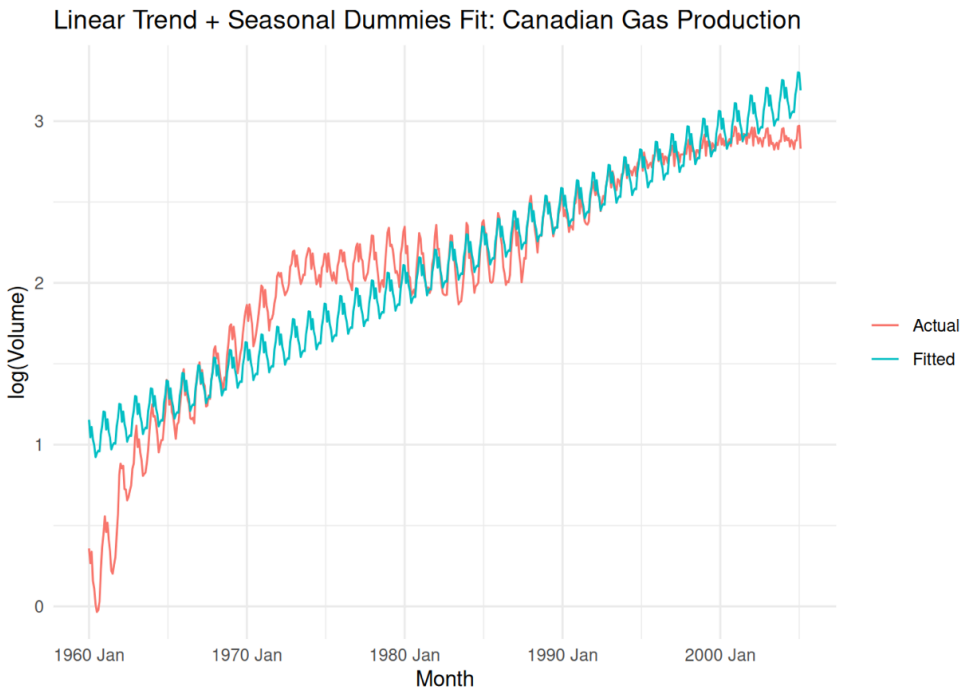
term	estimate	std.error	statistic	p.value
(Intercept)	1.150	0.041	27.761	0.000
trend()	0.004	0.000	57.692	0.000
season()year2	-0.112	0.052	-2.144	0.032
season()year3	-0.052	0.053	-0.996	0.320
season()year4	-0.133	0.053	-2.537	0.011
season()year5	-0.174	0.053	-3.305	0.001
season()year6	-0.251	0.053	-4.777	0.000
season()year7	-0.230	0.053	-4.374	0.000
season()year8	-0.219	0.053	-4.170	0.000
season()year9	-0.227	0.053	-4.325	0.000
season()year10	-0.125	0.053	-2.383	0.018
season()year11	-0.078	0.053	-1.480	0.140
season()year12	0.007	0.053	0.137	0.891

```
can_gas_fit |>
  tidy() |>
  filter(term == "trend()") |>
  mutate(monthly_growth_pct = 100 * estimate) |>
  select(term, estimate, monthly_growth_pct) |>
  knitr::kable(digits = 4)
```

term	estimate	monthly_growth_pct
trend()	0.004	0.3973

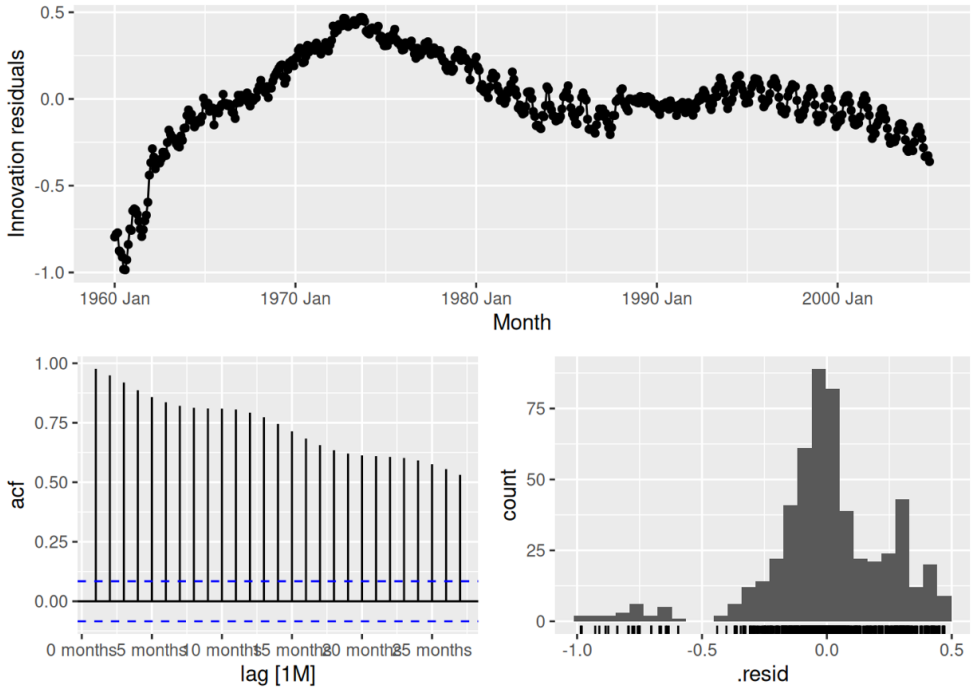
4.4a. The trend coefficient is about 0.004 so average monthly growth in gas production is about 0.4% per month. The seasonal dummies are measured proportionately to January. The most significant seasonal effect is June, with an estimate of -0.251 and a t-statistic of -4.78, the largest in absolute value. June is about 25 log points below January, which fits the winter-heating pattern. December and January are almost equal as December's coefficient is about 0.

```
can_gas_fit |>
  augment() |>
  ggplot(aes(x = Month)) +
  geom_line(aes(y = log_volume, color = "Actual")) +
  geom_line(aes(y = .fitted, color = "Fitted")) +
  labs(title = "Linear Trend + Seasonal Dummies Fit: Canadian G
       y = "log(Volume)", color = NULL) +
  theme_minimal()
```



4.4b. The fitted line captures the seasonal pattern well, with similar-sized yearly swings throughout. It misses the trend. The actual series is curved but the model forces a straight line. The fit overshoots in the early 60s, undershoots through the 70s and 80s, and overshoots again in 2000s.

```
can_gas_fit |> gg_tsresiduals()
```




```
can_gas_fit |>
  augment() |>
  features(.innov, ljung_box, lag = 24) |>
  knitr::kable(digits = 3)
```

.model	lb_stat	lb_pvalue
TSLM(log_volume ~ trend() + season())	7774.218	0

4.4c. The residuals do not look like white noise. The time plot shows long, slow swings and the ACF is very high, and decays slowly. The Ljung-Box test gives a statistic of about 7,774 with a p-value of basically 0, so at the 5% level we reject the null of white noise. A straight-line trend plus fixed seasonal dummies leaves a lot of predicable structure like the nonlinear trend and persistence.

Forecast Evaluation

Expanding-window TS-CV (5.1)

```
gas_cv_fc <- can_gas |>
  select(Month, Volume) |>
  stretch_tsibble(.init = 240, .step = 1) |>
  model(snaive = SNAIVE(Volume),
```

```
        tslm   = TSLM(log(Volume) ~ trend() + season())) |>
  forecast(h = 1)

cv_expanding_acc <- gas_cv_fc |>
  accuracy(can_gas, measures = list(MAE = MAE, RMSE = RMSE))

cv_expanding_acc |>
  select(.model, MAE, RMSE) |>
  knitr::kable(digits = 3,
               caption = "Expanding-window TS-CV accuracy (one-
```

Expanding-window TS-CV accuracy (one-month-ahead)

.model	MAE	RMSE
snaive	0.562	0.702
tslm	4.009	4.362

```
gas_cv_fc |>
  as_tibble() |>
  filter(Month <= yearmonth("2005 Feb")) |>
  summarize(first_forecast = min(Month), last_forecast = max(Mo
  knitr::kable()
```

first_forecast	last_forecast
1980 Jan	2005 Feb

5.1a. The first forecast is Jan 1980 and last evaluated forecast is Feb 2005 which is 302 forecasts. The seasonal random walk forecasts far more accurately than the trend + seasonal dummies model. Its errors are roughly seven times smaller and both metrics agree.

Is the difference significant? (5.2)

```
errs <- gas_cv_fc |>
  as_tibble() |>
  left_join(as_tibble(can_gas) |> select(Month, actual = Volume
            by = "Month") |>
  mutate(e = actual - .mean) |>
  select(Month, .model, e) |>
  pivot_wider(names_from = .model, values_from = e) |>
  na.omit()

forecast::dm.test(errs$snaive, errs$tslm, h = 1, power = 1)
```

```
    Diebold-Mariano Test

data:  errs$snaiveerrs$tslm
DM = -31.278, Forecast horizon = 1, Loss function power = 1,
p-value <
2.2e-16
alternative hypothesis: two.sided
```

5.2a. The DM statistic is -31.28 with a p-value below 2.2e-16. We reject the null that the two models are equally accurate at any usual significance level. The negative sign means the first model (SNAIVE) has smaller average absolute errors so the SNAIVE forecasts are significantly better than the TSLM forecasts.

Rolling window (5.3)

```
gas_cv_fc_roll <- can_gas |>
  select(Month, Volume) |>
  slide_tsibble(.size = 240, .step = 1) |>
  model(snaive = SNAIVE(Volume),
        tslm   = TSLM(log(Volume) ~ trend() + season())) |>
  forecast(h = 1)

cv_rolling_acc <- gas_cv_fc_roll |>
  accuracy(can_gas, measures = list(MAE = MAE, RMSE = RMSE))

cv_rolling_acc |>
  select(.model, MAE, RMSE) |>
  knitr::kable(digits = 3,
               caption = "Rolling-window (240 months) TS-CV acc
```

Rolling-window (240 months) TS-CV accuracy (one-month-ahead)

.model	MAE	RMSE
snaive	0.562	0.702
tslm	2.095	2.425

5.3a. SNAIVE didn't change at all because it only uses the same month last year, so a longer or shorter training window doesn't matter. TSLM improved a lot (MAE from 4.01 to 2.09), because it relies on a straight-line trend estimated from the whole training window. As we saw in 4.4b, a single straight line fits this series poorly since growth was steep early on and slowed later. The expanding window keeps the old steep-growth data in every fit, while the rolling window drops it and tracks recent behavior

better. TSLM is still worse than SNAIVE though.

Reflection

The residuals from the simple trend + seasonal dummies model still showed a lot of structure: long, slow swings and an ACF near 0.9 at the first lag, which means the model missed the changing, nonlinear trend and the persistence in the series. The Ljung-Box test confirmed this by rejecting white noise confidently. The forecast evaluation showed the cost of that missing structure: the seasonal random walk beat the trend + seasonal model by a wide margin in both the expanding and rolling windows, and the DM test says the gap is statistically significant. The rolling window helped the trend model because it dropped old data from a very different growth period, which suggests the trend is changing over time. I would expect a better model to let the level of the series adapt, for example by including the series' own past values or a trend that can change, while keeping the seasonal pattern. This model's residuals should look much closer to white noise, and its forecasts should be able to beat the seasonal random walk.